

Reproducible Design-Space Study of Lightweight Lattice-Based Authentication and Signatures for Blockchain Transactions

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Abstract

Quantum computers threaten the classical public-key primitives (RSA, ECDSA) used by most blockchains today; NIST has therefore standardized lattice-based signatures—ML-DSA (FIPS 204) and FN-DSA (a FALCON-based scheme being finalized as FIPS 206)—and mandated a transition of the underlying cryptosystems. This report describes a *reproducible, teaching-level* study of the design space of lightweight lattice-based authentication and signatures for blockchain-style transactions. We study two constructions: (i) Track A, a symmetric LWE authentication tag whose matrix is derived per transaction from SHA-256 (“one transaction, one matrix”), and (ii) Track B, a ringless public-key lattice signature in the Lyubashevsky Fiat–Shamir-with-Abort style. Our contribution is methodological rather than a claim of a new provably secure scheme: we (a) propose a parameter instantiation workflow that combines a root-Hermite/core-SVP estimator with scaled-down LLL embedding attacks and validate the two layers against each other; (b) study a message-derived matrix variant that removes the need to transmit a matrix seed; and (c) release an open, fully reproducible benchmark (E1–E4) that follows recent methodological critiques of ML-DSA evaluation by reporting full latency distributions and rejection-sampling worst cases instead of means only. All code, data, seeds, and figures are public. We do *not* claim formal security proofs; formal analysis and module/ring-structured instantiations are left to future work.

1 Introduction

A growing portion of public ledgers relies on RSA and ECDSA for transaction authentication. Shor’s algorithm shows that both are broken in polynomial time on a scalable quantum computer, and NIST has standardized the lattice-based replacements—ML-DSA (FIPS 204) [10] and FN-DSA, the FALCON-based scheme being finalized as FIPS 206 [5]—with an explicit recommendation to phase out classical ECC in the near term [10]. Recent surveys of post-quantum migration for blockchains systematically report that lattice signatures inflate transaction and block sizes (order 10–30× versus ECDSA) and shift the verification cost profile [7, 6]. In this setting, the concrete engineering question is not only “which standard scheme” but “how do size, latency, and the sampling behavior of signature algorithms trade off at a given security level.” Answering that question carefully requires a benchmarking methodology that is currently an active research topic in its own right [11].

This work studies two lightweight constructions with the semantics of the blockchain use case kept explicit:

- **Track A — symmetric LWE authentication tag (MAC).** The issuer derives an $n \times n$ matrix A deterministically from the transaction payload, computes a tag $b = As + e \pmod{q}$ with a short ternary secret s , and the key-holding verifier checks $\|b - As\|_\infty \leq V$. Because verification requires the secret s , Track A has *MAC semantics*; it targets the

“broadcast to a designated offline verifier” scenario and is not directly comparable with public-key signatures.

- **Track B — publicly verifiable lattice signature.** We implement a ringless signature in the Lyubashevsky Fiat–Shamir-with-Abort style [8, 9]: the public key is $(A, T = AS)$, the signer masks with a uniform vector and rejects until the released vector is short, and anyone can verify without a shared secret.

Our contributions are:

1. **I1 — parameter methodology with cross-validation.** A teaching-level security estimator based on the root-Hermite factor and core-SVP costs is used to instantiate parameters at a target security level. Because full-size lattice attacks are computationally out of reach on a laptop, we additionally run *scaled-down* LLL embedding attacks at $q = 257$ and verify that the empirically observed key-recovery rate moves in the direction predicted by the estimator as dimension, noise, and sample count vary. The two layers of evidence (estimate + phase-transition experiment) are meant to be used together; neither is a certificate. A reference lattice-estimator cross-check (Section 4.3) quantifies how far the teaching estimate can overshoot in the saturated regime and supplies corrected size guidance.
2. **I2 — transaction-derived matrix.** Instead of storing or transmitting a matrix seed in the public material, the matrix A is expanded from $\text{SHA-256}(tx)$ with domain separation. This eliminates the seed from the transmitted credential/signature and prevents cross-transaction reuse of the same instance. We explicitly flag the security caveat: a message-dependent A lets an adversary choose matrices adversarially, which requires an additional reduction argument that is out of scope here.
3. **I3 — open reproducible benchmark.** We publish the evaluation protocol, seeds, machine metadata, per-sample timing data, rejection-sampling histograms, block-inflation estimates, and a one-command reproduction pipeline, aligned with the evaluation practices recommended by the recent ML-DSA benchmarking critique [11].

Scope and honesty statement. This report is a *design-space and reproducibility study*, not an announcement of a new cryptosystem. All security figures are teaching-level estimates of relative hardness trends. We deliberately do not report the “defense-rate” comparisons of our earlier prototype (a strawman “LWE 100% vs. RSA 0%” framing), because those two attacks were not semantically comparable; the discarded material is kept in the repository only for provenance.

2 Preliminaries and Related Work

2.1 LWE and SIS

For a prime q , the *Learning With Errors* (LWE) problem asks to recover a secret $s \in \mathbb{Z}_q^n$ given m samples $(a_i, b_i = \langle a_i, s \rangle + e_i \pmod{q})$ where e_i is drawn from a (discrete) Gaussian of width σ . The *Short Integer Solution* (SIS) problem asks for a nonzero short vector x in the kernel lattice of a random $A \in \mathbb{Z}_q^{m \times n}$. Both problems underlie the NIST lattice standards. We use a ternary secret in $\{-1, 0, 1\}^n$ for Track A and a ternary secret matrix in Track B; the security estimator accounts for the reduced secret entropy as variance.

2.2 Fiat–Shamir with Aborts

Lyubashevsky introduced rejection sampling into the Fiat–Shamir transform so that the released response $z = y + Sc$ is statistically independent of the secret S [8], and later removed trapdoors from lattice signatures entirely [9]. This is the framework used by CRYSTALS-Dilithium/ML-DSA. Recent academic work continues to compress signatures and reduce sampling cost in this framework: HAETA uses bimodal rejection sampling [3], G+G removes rejection sampling via convolved Gaussians [4], Rhyme proposes a 3C sampling that avoids flooding, rejection sampling, and Gaussian convolution [12], and a very recent line achieves the first tight reductions from *search-LWE* without trapdoors [2]. Our Track B is a deliberately simple, ringless reference implementation in the same Fiat–Shamir-with-Aborts family; it is a baseline for methodology, not a competitor to these optimized schemes.

2.3 Benchmarking methodology

Riou et al. show that ML-DSA signing time is highly variable because of rejection sampling, and that reporting only means systematically underestimates worst-case execution time by up to roughly $20\times$ [11]. They recommend fixed input sets, per-sample timing, and reporting quantiles and worst cases rather than single-point averages. We follow that protocol in E2/E4 and additionally release the raw samples so that empirical CDFs can be recomputed.

2.4 Blockchain migration surveys

Surveys of post-quantum blockchains consistently identify signature size as the dominant driver of ledger inflation and quantify verification overhead against the ECDSA baseline [7, 6]. We use their reporting conventions (bytes, block-inflation multiple, verify TPS) in the blockchain scenario experiment E4.

3 Constructions and Threat Models

All arithmetic is over $\mathbb{Z}_q$ with a prime $q = 12289$ (12-bit modulus used by many lattice instantiations) unless stated otherwise. Deterministic expansion uses SHA-256 with domain separation and a counter-mode PRG; the implementation is documented in the repository.

3.1 Track A: symmetric LWE authentication tag

Parameters: n , q , σ (Gaussian width), tail t , and verification tolerance V . Setup chooses a uniform ternary secret $s \in \{-1, 0, 1\}^n$. For a transaction tx :

1. **Derive** $A = \text{Expand}(\text{SHA-256}(\text{tag-domain}||tx)) \in \mathbb{Z}_q^{n \times n}$;
2. **Issue** $b = As + e \pmod{q}$, where each e_i is drawn from a truncated discrete Gaussian of width σ and support $\{-t, \dots, t\}$;
3. **Verify** (key holder only): recompute A from tx and accept iff $\|b - As\|_\infty \leq V$.

Because the verifier needs s , Track A is a *message authentication tag*, not a digital signature. It is appropriate for “broadcast to a designated verifier with an established shared key”; it cannot provide public verifiability or non-repudiation. Its adversary goal is existential forgeability of a fresh tag from observed (tx_i, b_i) transcripts, i.e., a *search-LWE* instance over the collected samples.

Table 1: Semantics of the two constructions studied in this paper.

Scheme	Semantics	Verifier	Attack model
Track A (tag)	MAC (symmetric)	key holder only	forge fresh tag from (tx_i, b_i) transcripts
Track B (sig.)	digital signature	any party	find short S' with $AS' = T$
RSA/ML-DSA/FN-DSA	digital signature	any party	standard baselines

3.2 Track B: publicly verifiable lattice signature

Parameters: $A \in \mathbb{Z}_q^{m_a \times n}$ with $m_a = 224$ rows and $n = 256$ dimensions, secret $S \in \{-1, 0, 1\}^{n \times m_c}$ with $m_c = 256$ (each entry nonzero with probability $2/3$), challenge length $m_c = 256$ and Hamming weight $h = 32$, mask radius κ , acceptance bound $\kappa - h$, and hash $H = \text{SHA-256}$. The public key is $(A, T = AS \bmod q)$. Keygen, sign, and verify are:

1. **Keygen:** sample A uniformly from a seed, sample the ternary secret matrix S , and set $T = AS \bmod q$.
2. **Sign**(tx): repeat: (1) sample $y \in [-\kappa, \kappa]^n$ uniformly and compute $w = Ay \bmod q$; (2) compute the digest $c = H(tx \parallel \text{encode}(w))$ and expand it into a sparse ternary challenge vector with Hamming weight h ; (3) set $z = y + Sc$ and *accept* only if $\|z\|_\infty \leq \kappa - h$; otherwise restart. Output the signature (z, digest) .
3. **Verify**($tx, (z, \text{digest})$): reject if $\|z\|_\infty > \kappa - h$; recompute c from the digest; accept iff $H(tx \parallel \text{encode}(Az - Tc)) = \text{digest}$.

Correctness follows from $Az - Tc = Ay + AS c - AS c = w \bmod q$. Conditioned on acceptance, $z = y + Sc$ is uniformly distributed over $[-\kappa + h, \kappa - h]^n$ and is independent of S (statistical zero-knowledge), because y is uniform and the map $y \mapsto z$ is a bijection on the acceptance region. Forging a signature requires finding a short S' with $AS' = T \pmod{q}$, an *inhomogeneous* SIS instance. All of these statements are standard textbook arguments; we make no claim of new cryptanalysis. As discussed below, the ringless instantiation has a large public key (≈ 224 KB), which motivates module/ring structure in a production design.

3.3 Semantic boundaries

Keeping these boundaries explicit avoids the common trap of comparing a symmetric MAC against public-key signatures on “who prevents forgery better”; instead Track A is positioned for symmetric, key-holder-checked settings, and only Track B is placed on the same comparison plane as RSA-PSS and the NIST schemes.

4 Parameter Methodology (I1)

4.1 Teaching-level security estimator

For an LWE instance with n dimensions, m samples, modulus q , Gaussian width σ , and ternary secret variance $2/3$, we model the best known attack as a Kannan embedding into a lattice of dimension $d = m + n + 1$. The required root-Hermite factor is compared against the achievable factor of BKZ- β ,

$$\delta(\beta) = \left(\frac{\beta}{2\pi e} (\pi\beta)^{1/\beta} \right)^{1/(2(\beta-1))},$$

and the core-SVP cost is taken as $2^{0.292\beta}$ (classical) or $2^{0.265\beta}$ (quantum), following the NewHope cost model. For a target of ≥ 80 bits this scan returns $n = m = 144$, $q = 12289$, $\sigma = 2$, tail

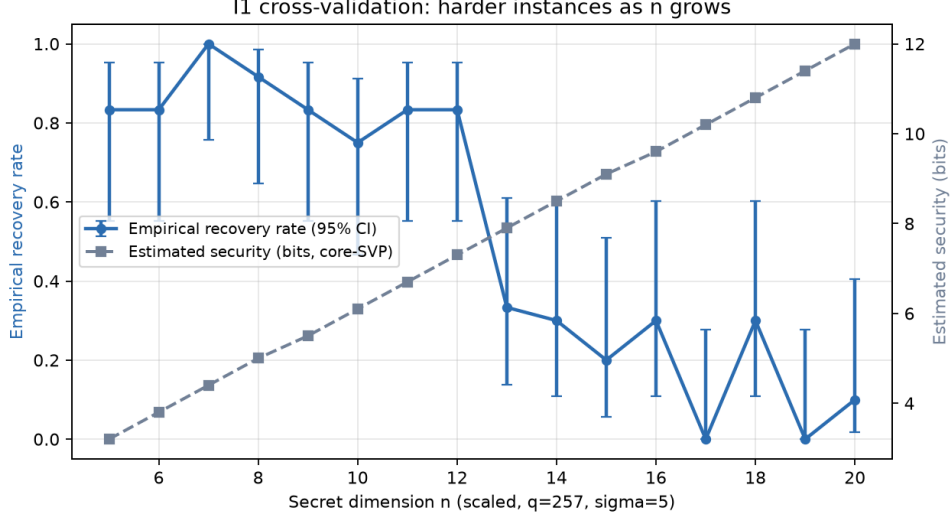


Figure 1: I1 cross-validation (scaled, $q = 257$, $\sigma = 5$): the empirical key-recovery rate of a Kannan-embedding LLL attack (left axis, 95% Wilson CI) decreases as the secret dimension n grows, while the teaching estimator’s bit-security (right axis) increases. Each point: 8–12 independent instances.

$t = 6$, $V = 8$, with an estimate of 84.4 bits at full-dimension block size $\beta = 289$. As quantified in Section 4.3, this figure is the estimator’s *dimensional-saturation ceiling*, not a supported security estimate: the reference lattice-estimator puts the same instance at roughly 2^{12} – 2^{41} of attack cost. For Track B we check the SIS regime: with $n = 256$ dimensions and $m_a = 224$ rows the target norm lies about -30.5 dB below the Gaussian heuristic bound, i.e., in the region where finding a short solution requires lattice reduction.

4.2 Scaled-down cross-validation

Directly attacking the full-size parameters is out of reach, so we validate trends on scaled instances with $q = 257$ and $\sigma = 5$ so that the target/GH ratio is near 1 and phase transitions are observable in small dimensions. Our attack is a self-contained Kannan embedding using a QR-accelerated floating point LLL (dimensions ≤ 41), followed by Babai nearest-plane decoding; each parameter point uses 8–12 independent random instances with a 120s time budget per point. We sweep (a) secret dimension $n \in \{5, \dots, 20\}$, (b) the number of historical samples $M \in \{1, \dots, 6\}$ at fixed $n = 6$, and (c) the noise width $\sigma \in \{1, \dots, 7\}$ at fixed $n = 12$. Section 7 reports that the measured success rates move *in the direction* predicted by the estimator (harder as n grows or σ grows; easier as M grows). This is a consistency check of the estimation *trends*, not a security proof for any concrete parameter set. Section 4.3 reports a reference lattice-estimator cross-check that quantifies how far the teaching model can overshoot in the saturated regime and gives corrected size recommendations.

4.3 Reference cross-check with lattice-estimator

To obtain an independent reference point, we evaluate the Track A and Track B instances with the community lattice-estimator [1] (git commit 53da598, 2026-08-19). The tool is normally distributed as a SageMath package; to keep this artifact dependency-light we executed its attack code *unmodified* under CPython through a small pure-Python layer that re-implements the handful of `sage.all` numerical symbols the package imports (53-bit reals with an MPFR-scale exponent range, integer and rational rings, binomial/log/erf, and the normal/ χ^2 CDFs). The layer is validated against the estimator’s own doctest reference values before any new numbers

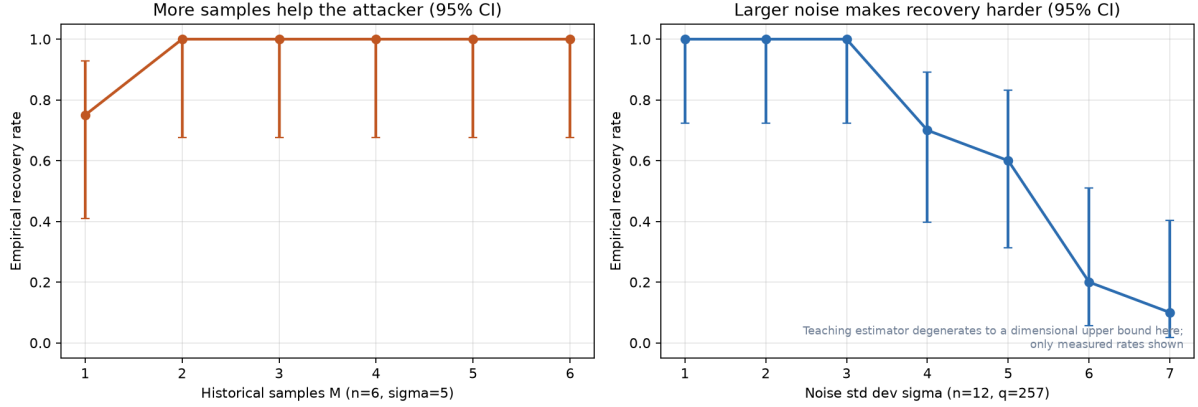


Figure 2: Left: recovery rate vs. the number of historical LWE samples M ($n = 6$, $\sigma = 5$); more samples help the attacker. Right: recovery rate vs. noise width σ ($n = 12$); larger noise makes recovery harder. In this low-dimensional teaching regime the estimator degenerates to a dimensional bound, so only measured rates (with 95% Wilson CI) are shown.

are produced; Table 2 shows that every reference row reproduces exactly at the printed precision. We report four attack families: primal uSVP and BDD under the estimator’s default reduction-cost model, and uSVP plus dual-hybrid under the ADPS16 core-SVP / GSA model used by `estimate.rough` (the literature-comparable numbers). We do not report the default-model full dual family: that routine internally amplifies success probabilities below the IEEE-754 range, where only Sage’s MPFR semantics apply; its conservative core-SVP analogue (rough dual-hybrid) is included instead.

Table 2: Validation of the numerical layer. Reference values are the lattice-estimator doctest outputs; “this work” values come from identical calls in this artifact. All rows match at the printed precision (± 0.1 bit).

Instance	Call	ref. cost ($\log_2$)	ref. β / d	this work
Kyber512	uSVP (default)	143.8	406 / 998	same
Kyber512	BDD (default)	140.2	389 / 1005	same
Kyber512	uSVP (ADPS16/GSA)	118.6	406 / 998	same
Dilithium2 MSIS	SIS lattice (rough)	123.5	423 / 2303	same

Track A. Table 3 reports the reference costs for the benchmarked prototype ($n = m = 144$) and for $n = m = 480$, the smallest size in our scan at which the conservative ADPS16 uSVP estimate reaches ≈ 91 bits. Under the reference estimator the $n = 144$ prototype is worth only $\approx 2^{41}$ (default uSVP/BDD) to $\approx 2^{12}$ (core-SVP uSVP) word operations—far below the 84.4 bits of the teaching scan, which saturates at its dimensional ceiling ($\beta = d = 289$). The teaching estimator is therefore a *loose upper bound* in this regime, not a security estimate, and the agreement seen in E3 is a validation of *trends* rather than of absolute cost. The default-model scan reaches 80 bits at $n \approx 320$, but the conservative core-SVP rows require $n \approx 440$ –480; we therefore recommend $n = m = 480$, $q = 12289$, $\sigma = 2$ as a ~ 90 bit teaching instantiation of Track A (Figure 3). All E1–E4 benchmark data in this report keep the original $n = 144$ prototype and are methodology-level, not security-level, data.

Track B. We cross-checked the SIS core used by the methodology, namely the homogeneous kernel of the 224×256 matrix A with the Euclidean bound $\sqrt{256 \cdot 2/3} \approx 13.1$ of a planted ternary column. The reference lattice attack reports *no* admissible vector (cost 2^∞ , $\beta = d = 224$):

Table 3: Reference lattice-estimator costs for the Track A LWE-tag family ($q = 12289$, $\sigma = 2$, independent ternary secret, $m = n$). Values are $\log_2$ word operations of the cheapest reported row per family.

n	uSVP (default)	BDD (default)	uSVP (ADPS16)	dual-hybrid (ADPS16)
144 (prototype)	41.3	41.1	12.0	17.7
480 (recommended)	117.9	114.8	91.4	89.4

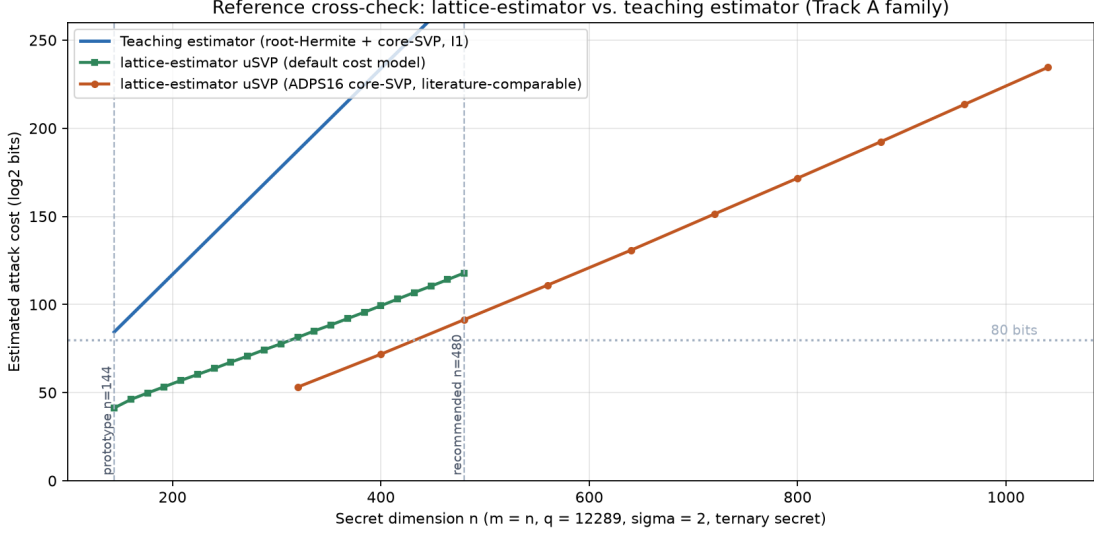


Figure 3: Reference cross-check on the Track A family. The teaching root-Hermite/core-SVP estimator (blue) grows linearly with the secret dimension because it saturates at block size $\beta = d = 2n + 1$; the reference lattice-estimator uSVP estimates under the default model (green) and under the ADPS16 core-SVP model (orange) are far lower at the benchmarked $n = 144$. A prototype of $n = 480$ clears ~ 80 bits under both reference models.

for a uniformly random matrix the Gaussian-heuristic shortest vector of this q -ary lattice is $\lambda_1 \approx 1.5 \times 10^4$, so no 13-bounded kernel vector exists in the generic model. This confirms that the relevant hardness of Track B is the *planted, inhomogeneous* (preimage) problem, which the reference tool does not model directly, and it explains why Track B is reported “in the SIS regime” without a claimed bit count. Two design constraints follow for future parameter derivation. First, the infinity-norm MSIS attack in the reference tool requires the length bound to stay below $(q - 1)/2 = 6144$; the two-signature relation of Section 3.2 can have coordinates up to $2(\kappa - h) = 7848$, so parameter selection must either reduce $\kappa - h$ or increase q until $2(\kappa - h) < (q - 1)/2$. Second, a bit-level instantiation of the inhomogeneous instance (e.g. via module structure as in ML-DSA) is left to future work.

5 Transaction-Derived Matrix (I2)

Lattice schemes such as ML-DSA expand a short random *seed* into the matrix A so that the public key carries only the seed. I2 applies the same idea at the message level: for Track A (and similarly for Track B challenges), we derive the matrix from the transaction hash with a domain-separation prefix,

$$A_{tx} = \text{Expand}(\text{SHA-256}(\text{domain}||tx)).$$

This has two engineering consequences. First, no matrix seed needs to be transmitted or stored alongside the credential, since every verifier can recompute A_{tx} ; the tag/signature remains

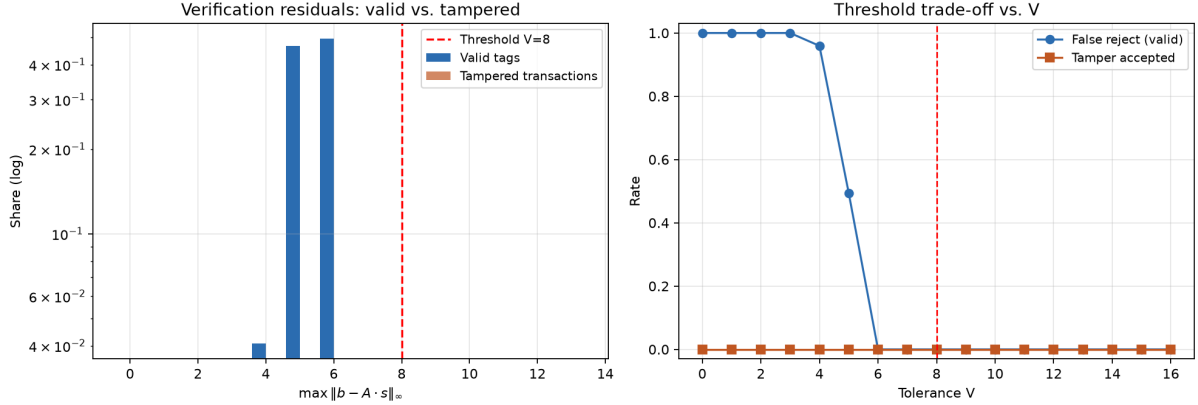


Figure 4: E1 (Track A). Left: normalized histogram (log scale) of verification residuals $\|b - As\|_\infty$ for valid tags and for tampered transactions ($V=8$); residuals of tampered transactions exceed the histogram range shown here. Right: false-reject rate (valid) and tamper-pass rate vs. tolerance V .

compact. Second, because A changes with the transaction, an adversary cannot amortize one instance across many signed messages; E3(b) confirms empirically that accumulating samples from the same instance makes key recovery markedly easier (Section 7), which is exactly the reuse that per-transaction derivation prevents.

Security caveat. Making A a function of the message gives the adversary some control over the matrix by choosing messages, so the reduction must account for adversarially chosen A . A full treatment (e.g., randomizing the derivation or proving hardness under a suitable variant) is future work. We therefore present I2 as an engineering feasibility study with a clearly stated boundary, not as a proven construction.

6 Experimental Evaluation

6.1 Environment and protocol

All experiments run single-threaded on a Windows 11 desktop (AMD desktop CPU, Family 25 Model 68, Python 3.12.10, NumPy 2.5.3, matplotlib 3.11.1, OpenSSL 3.5.7), with machine metadata embedded in the result JSON files. Timing uses `time.perf_counter` and includes Python call overhead; the RSA baseline is measured with OpenSSL’s in-process `openssl speed` (industry-standard implementation), and per-sample distributions are unavailable there, so it is marked as constant. All random seeds are fixed (E1: 12345; E1b/E2: 20260908; E3: 20260908) and the entire pipeline runs with `python scripts/run_v2_all.py`.

6.2 E1: correctness and threshold behavior (Track A and Track B)

For $n = 144$, $q = 12289$, $\sigma = 2$, $V = 8$, we test 1500 valid transactions, 1500 bit-flipped tampered transactions, and 300 random blind tags. All 1500 valid tags are accepted, all 1500 tampered transactions are rejected, and all 300 blind tags are rejected (each rate = 1.0000). Figure 4 (left) shows that verification residuals of valid tags concentrate in $[5, 6]$ (mean 5.45, max 6) while tampered transactions produce residuals far above the threshold. The right panel sweeps $V \in \{0, \dots, 16\}$: the false-reject rate is 1.0 for $V \leq 3$, 0.959 at $V = 4$, 0.494 at $V = 5$, and 0 for $V \geq 6$, while the tamper-pass rate is 0 across the entire sweep. Choosing $V = 8$ leaves a comfortable margin below the empirical residual maximum of 6.

Table 4: Size and timing comparison. Rows for Track A/B and RSA-3072-PSS are measured on this machine; the NIST rows use official sizes—ML-DSA-44 (FIPS 204) and FN-DSA-512 (FALCON v1.2, the scheme planned as FIPS 206)—and are not measured here. Security levels are teaching-level and not matched across rows. Signing medians with p99 in parentheses; verify median. Sig. = signature or credential.

Scheme	pk (B)	sk (B)	sig/cred (B)	sign med (ms)	verify med (ms)
Track A tag (symmetric)	—	144	288	2.34 (p99 4.99)	0.098
Track B signature (ringless)	229 376	65 536	544	1.36 (p99 9.66)	0.160
RSA-3072-PSS (OpenSSL 3.5.7)	384	384	384	7.39	0.135
ML-DSA-44 (official)	1312	2560	2420	ref.	ref.
FN-DSA-512 (FALCON)	897	1281	666	ref.	ref.
ECDSA P-256 (official)	65	32	64	ref.	ref.

Track B correctness. At the prototype scale used in E2 ($n = 256$, $m_a = 224$, $h = 32$, $\kappa = 3956$), we additionally test Track B in four groups of 1000 trials: (i) all valid signatures verify; (ii) all bit-flipped transactions are rejected; (iii) all random forgeries—uniform random z with $\|z\|_\infty \leq \kappa - h$ and a fresh 32-byte digest—are rejected; and (iv) all signatures mutated in a single byte are rejected (each rate = 1.0000). Unit tests in the repository cover the same round-trip and tamper paths at smaller scale. Result JSON: `data/eval_v2_e1b.json`.

6.3 E3: scaled attacks (I1 cross-validation)

Figure 1 sweeps the secret dimension n at $q = 257$, $\sigma = 5$: the measured recovery rate falls from 0.83 at $n = 5$ to 0.10 at $n = 20$ (per-point 95% Wilson intervals; small-sample noise between adjacent points is expected), while the estimator rises from 3.2 to 12.0 estimated bits. Figure 2 (left) fixes $n = 6$ and varies the number of samples: recovery rises from 0.75 ($M = 1$) to 1.0 ($M \geq 2$), confirming that accumulating samples from the *same* instance favors the attacker. Figure 2 (right) fixes $n = 12$ and varies σ : recovery drops from 1.0 ($\sigma \leq 3$) to 0.1 ($\sigma = 7$). The qualitative agreement between the empirical phase transitions and the estimator is the central evidence for I1.

6.4 E2: prototype size and latency comparison

Table 4 summarizes sizes and latencies. All rows use the teaching-level instantiations of Section 4 (Track A: teaching estimate 84.4 bits, but the reference cross-check of Section 4.3 puts the $n = 144$ prototype at roughly 2^{12} – 2^{41} ; Track B: reported in the SIS regime without a claimed bit count; RSA-3072: ≈ 112 bits), so the rows are not at a matched security level. Track B signing has median 1.36 ms but a heavy tail ($p_{90} = 4.49$ ms, $p_{99} = 9.66$ ms, max 24.1 ms) driven by rejection sampling; the number of retries per signature has median 6 and reaches 61 in the worst case. Figure 5 (left) shows the empirical CDFs of signing time, where the Track B long tail is clearly visible against the near-constant Track A and RSA values. Verification is fast for all schemes (≤ 0.16 ms median here). Track A “signing” (issuing the tag) has median 2.34 ms and p99 4.99 ms; its low p99/max spread reflects the absence of rejection sampling.

6.5 E4: blockchain ledger scenario

We simulate a single node receiving a block of $N = 1000$ transactions of 64 B payload each and verifying sequentially. The ledger inflation factor is the ratio of signature overhead to payload; verification TPS is derived from the E2 median verify times. Table 5 reports the results. Track A adds 4.5 $\times$ and Track B 8.5 $\times$ overhead at measured verification throughput of about 10 000 and 6 300 TPS respectively, both faster than the measured RSA-3072-PSS signing-oriented baseline

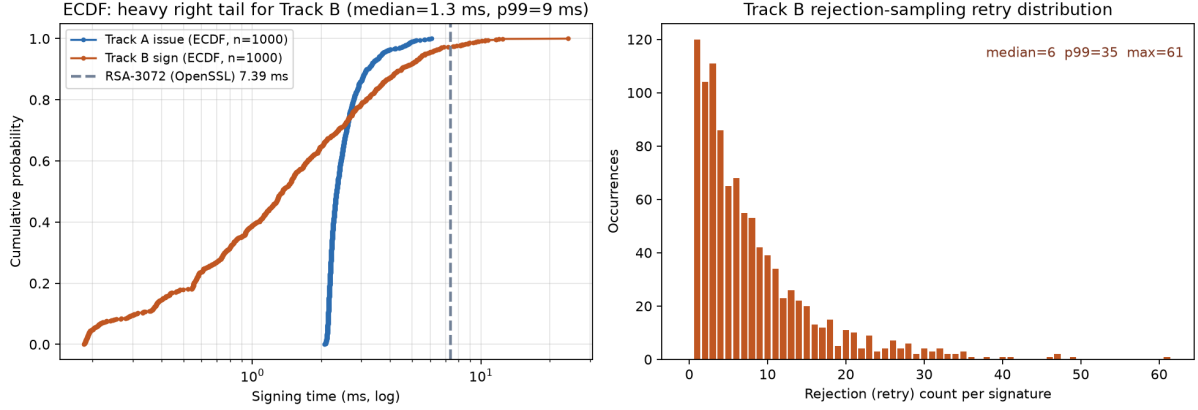


Figure 5: E2. Left: empirical CDF of signing time (log axis) for Track A issue and Track B sign ($n=1000$ each) with the constant RSA-3072 value for reference; the Track B heavy tail is visible (median 1.3 ms, p99 9 ms). Right: histogram of Track B rejection-retry counts (median 6, p99 35, max 61).

Table 5: E4: per-transaction overhead and single-node verification throughput for a block of 1000 transactions of 64 B each. Rows marked (ref.) use official spec sizes (FN-DSA-512: FALCON v1.2 / FN-DSA draft) and literature-order times, not measurements on this machine.

Scheme	Inflation vs. payload	Verify TPS (median)
Track A tag (measured)	$4.5\times$	10 235
Track B signature (measured)	$8.5\times$	6 262
RSA-3072-PSS (measured)	$6.0\times$	7 407
ML-DSA-44 (ref.)	$37.8\times$	$\sim 20\,000$
FN-DSA-512 (ref.)	$10.4\times$	$\sim 12\,500$
ECDSA P-256 (ref.)	$1.0\times$	$\sim 100\,000$

on this machine. For the NIST reference rows we use official signature sizes (FN-DSA-512: FALCON v1.2 / FN-DSA draft values) and literature-order verification costs (not measured here): ML-DSA-44 inflates by $37.8\times$, FN-DSA-512 by $10.4\times$, and ECDSA P-256 is the $1.0\times$ baseline. These orders of magnitude reproduce the pattern reported by post-quantum migration surveys [7, 6]: signature size, not verification arithmetic, dominates ledger growth.

7 Results Summary and Discussion

The experiments support four high-level observations.

- **Correctness is clean at the tested scale.** Track A achieves 1.0000 on all three correctness groups, and the threshold sweep shows a wide operating margin ($V = 8$ sits above the empirical residual max of 6 and below any observed tamper residual). Track B likewise achieves 1.0000 across 1000 valid, 1000 tampered, 1000 random-forgery, and 1000 single-byte-mutation trials.
- **Estimator and scaled attack agree qualitatively.** Recovery rates fall when dimension or noise grows and rise when more same-instance samples accumulate, matching the estimator’s relative predictions. The agreement is trend-level; per-point noise (see the 95% CIs and non-monotonicities in Figure 1) is expected at 8–12 trials and should not be read as a precise “hardness curve.”

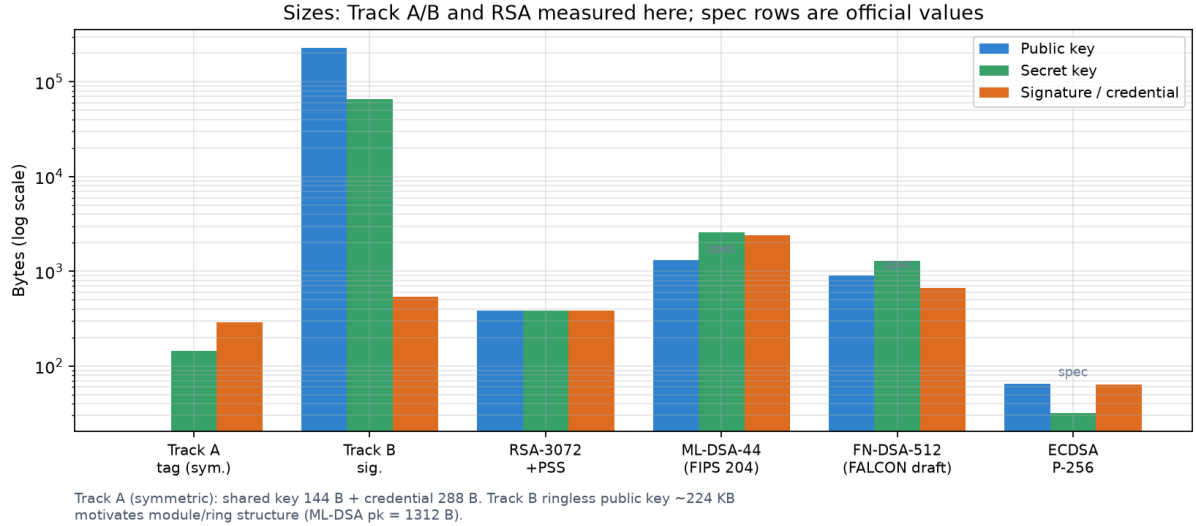


Figure 6: Public-key, secret-key, and signature/credential sizes (log axis). Track A/B and RSA rows are measured here; the NIST/FALCON rows are official spec values (FN-DSA-512 from FALCON v1.2). The ringless Track B public key (≈ 224 KB) is the engineering motivation for module/ring-structured instantiations such as ML-DSA (pk 1312 B).

- **Rejection sampling dominates the latency story.** The Track B signing distribution is long-tailed (median 1.36 ms, max 24.1 ms, worst case 61 retries), confirming the critique in [11] that mean-only reporting is misleading for designs with aborts. Any production deployment must budget for worst-case latency or bound retries via larger κ / better sampling.
- **Size, not arithmetic, is the blockchain bottleneck.** At 64-B payloads, signatures inflate blocks by 4.5–37.8 $\times$, and the measured verification throughput is in the thousands of TPS. The ringless Track B public key of ≈ 224 KB is the clearest sign that module/ring structure (as in ML-DSA) is required for real deployment.

7.1 Limitations

- **Teaching-level security estimates.** Our estimator is a root-Hermite/core-SVP model; in the regimes used here it often saturates at a dimensional upper bound and cannot distinguish fine effects of σ (as seen in Figure 2, right). The reference cross-check of Section 4.3 shows that in this saturated regime the teaching numbers overstate hardness by one to three orders of magnitude in bit terms, and we therefore treat them as trend indicators only. A complete production-level re-derivation (default dual family, bit-level Track B instance) remains future work.
- **Python + NumPy implementation.** All lattice code is in pure Python/NumPy on integer arrays; timings include interpreter overhead, and the LLL attack is a double-precision implementation valid only in small dimensions. Cross-language conclusions should rely on the OpenSSL/industry baselines and on the relative, not absolute, comparisons.
- **No formal proofs.** We present textbook-level correctness and zero-knowledge arguments; a rigorous ROM/QROM treatment, including the message-derived matrix caveat of I2, is future work.
- **Small attack trials.** E3 uses 8–12 instances per point under a time budget; the phase-transition observations are qualitative.

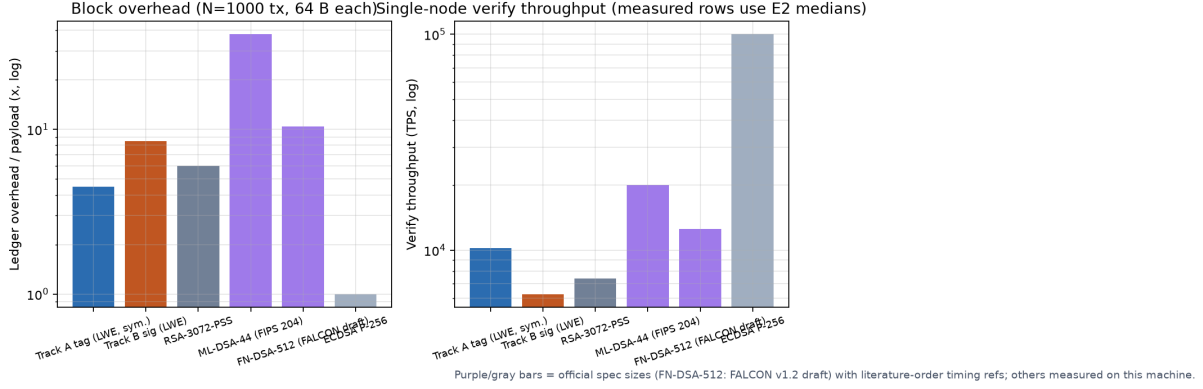


Figure 7: E4. Left: ledger-inflation multiple over the raw 64-B payload; right: single-node verification TPS (measured rows use E2 medians; purple/gray bars are official-size/literature-time references).

8 Conclusion and Future Work

We presented a reproducible design-space study of two lightweight lattice-based constructions for blockchain transactions, together with a parameter methodology that cross-validates a teaching-level estimator against scaled LLL embedding attacks, a transaction-derived matrix variant with explicit security caveats, and an open benchmark suite (E1–E4, plus a Track B correctness test) aligned with current methodological best practices. The results show that (i) the two layers of evidence agree in trend, (ii) rejection sampling creates a long tail that benchmarks must report as distributions, and (iii) signature/key size is the dominant blockchain cost driver, strongly motivating structured lattices.

Future work proceeds along four axes: (1) formal ROM/QROM security proofs, including a treatment of message-derived matrices; (2) module-LWE/SIS instantiations with NTT for compact keys and constant-time implementations; (3) a complete lattice-estimator re-derivation of production parameters, including the default dual-attack family and a bit-level Track B instance that respects the length-bound constraint of Section 4.3; and (4) batch verification and threshold-signature extensions (e.g., Ringtail-style protocols) with the same measurement protocol.

Reproducibility

- **Repository:** <https://github.com/linmcneil/quantum-resilient-signature> (public).
- **One-command pipeline:** `python scripts/run_v2_all.py` reproduces E1 (Track A), E1b (Track B correctness), E3, E2, E4, and all figures.
- **Seeds:** E1 = 12345 (per-instance derived seeds), E3 = 20260908, E2 = 20260908 with scheme instance seeds 11 (Track A) and 22 (Track B); machine metadata is embedded in each `data/eval_v2_*.json` (including `eval_v2_e1b.json`).
- **Artifacts:** raw per-sample timing (`samples_ms`), rejection-count histograms, threshold sweeps, and JSON result schemas are in `data/`; figures in `docs/figures_v2/` (Chinese) and `docs/figures_en/` (English).

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